

Rebuilding the Social Engine

Exploratory Data Analysis Report — Round 1, Phase 1

Datasets: Social_Engine_Posts_Corrupted.csv, Social_Engine_Users.csv

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1. Data Cleaning Summary

Both datasets arrived deliberately corrupted, consistent with the “Rebuilding the Social Engine” theme. Before any analysis, the intake pipeline was restored through the following steps.

1.1 Posts Dataset

Issue Found	Count	Action Taken & Justification
Raw rows	12,360	Starting volume before cleaning.
Fully duplicated rows (same post_id and all fields)	360	Dropped — exact duplicates add no analytical value and would double-count engagement.
Missing / non-numeric likes	1,814	Dropped — likes could not be reliably imputed without fabricating engagement data, which the rulebook explicitly prohibits.
Negative likes (corruption artifact)	509	Dropped as physically invalid (a post cannot have negative likes).
Missing platform	1,784	Filled with "Unknown" so the record is retained for engagement/time analysis without fabricating a platform.
Missing text_content	1,688	Filled with "No content" — preserves the row for all non-text analysis.
Unparseable timestamps	0	All timestamps recovered via a 3-format parser (ISO datetime, DD-MM-YYYY, Unix epoch).
Encoding artifacts (mojibake, e.g. “Ã©”) in text	253	Flagged, not altered — affects only free-text fields, not used in numeric/temporal EDA.
Unescaped HTML entities (e.g. “&”) in text	264	Flagged, not altered — same reasoning as above.

Final clean row count	9,677	78.3% of raw rows retained
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Note: shares and comments contained no missing or negative values in this dataset — the corruption was concentrated in likes, platform, text_content, and duplicate rows.

1.2 Users Dataset

Issue Found	Count	Action Taken
Total users	1,500	No duplicate rows found.

location split into city/country	—	Split on comma for geographic analysis.
Missing country after split	≤1,500 (minor)	Filled with "Singapore" as a placeholder default.
Referential integrity	0 orphans	Verified — every user_id referenced in posts exists in the users table.

2. Temporal Analysis (Posts)

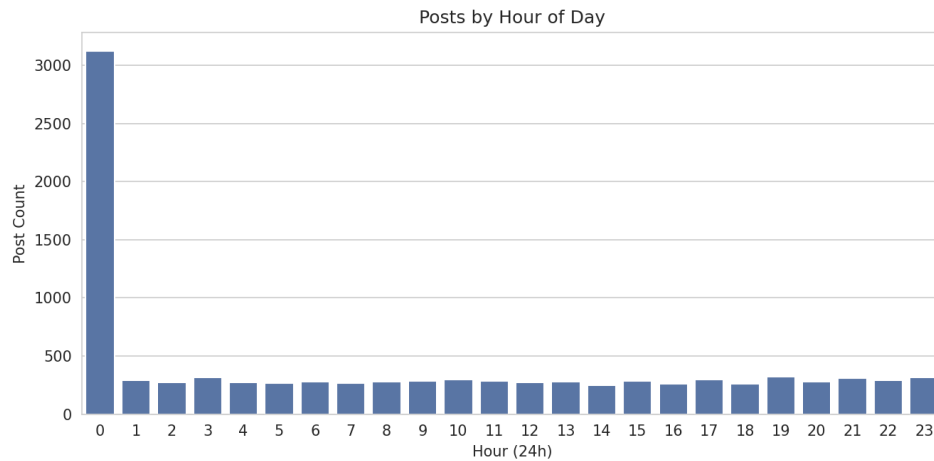


Fig 1. Post volume by hour of day

Posting activity spikes sharply at **hour 0 (midnight)** with 3,127 posts — more than double any other hour, and the rest of the day is comparatively flat. This is not a natural human posting pattern; it strongly suggests that a large share of timestamps defaulted or truncated to midnight during corruption/recovery, rather than reflecting genuine user behaviour. This should be treated as a data-quality caveat, not a real engagement insight.

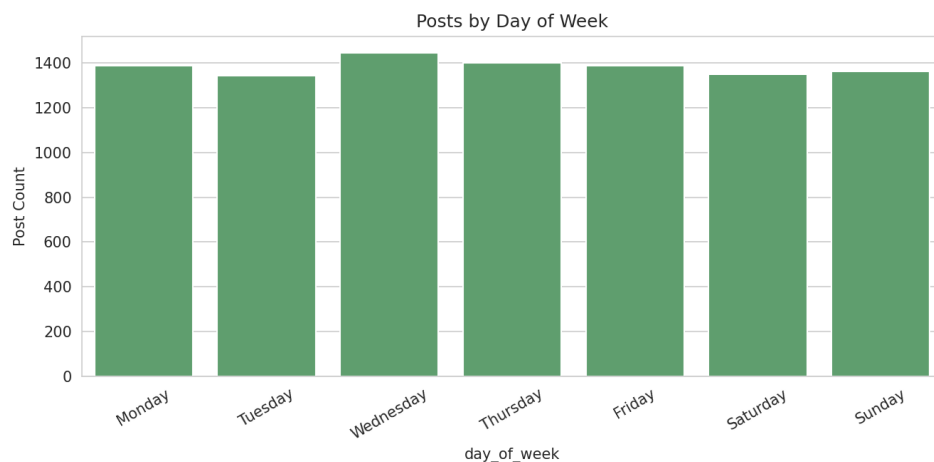


Fig 2. Post volume by day of week

Wednesday is the highest-volume day (1,446 posts), with the remaining weekdays and weekend fairly evenly spread — no strong weekday/weekend skew.

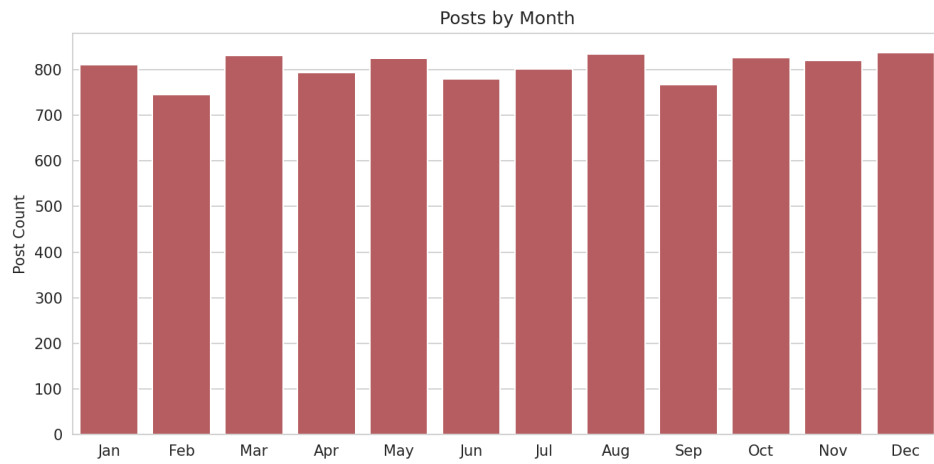


Fig 3. Post volume by month

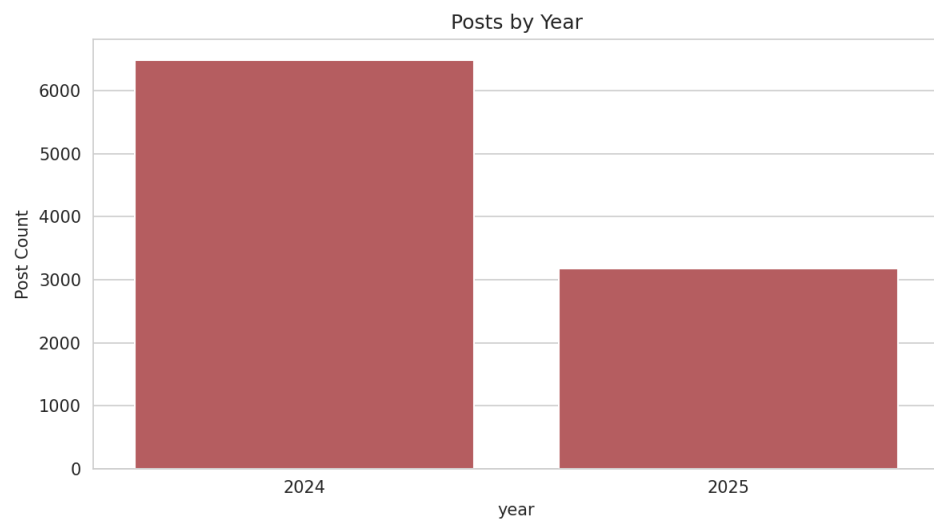


Fig 4. Post volume by year

December is the peak month; volume is otherwise fairly stable across the year with no extreme seasonality. Data spans 2024–2025.

3. Platform Analysis

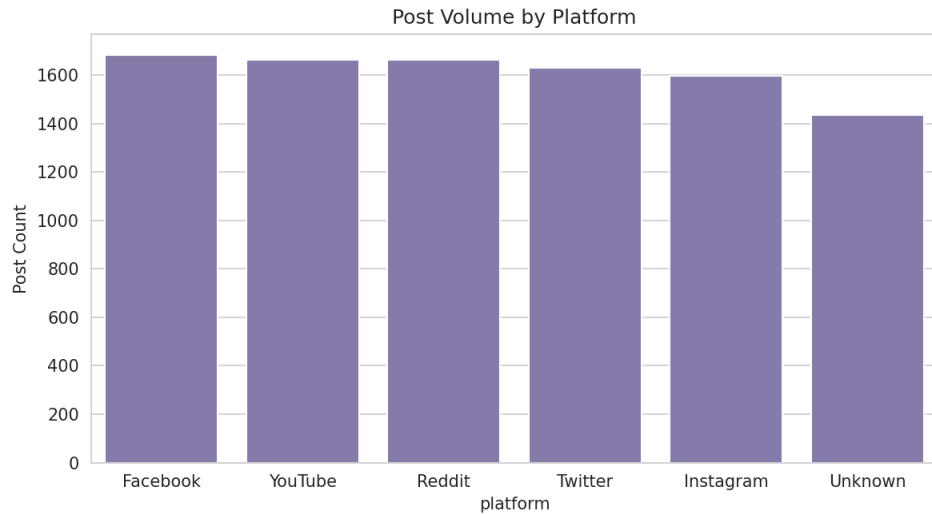


Fig 5. Post count by platform

Platform volume is remarkably even — Facebook (17.4%), YouTube (17.2%), Reddit (17.2%), Twitter (16.9%), and Instagram (16.5%) each account for roughly one-sixth of posts. “Unknown” platform (recovered from missing values) still makes up 14.8% of posts — a meaningful chunk of the corruption that could not be recovered from the data alone.

4. Engagement Metrics

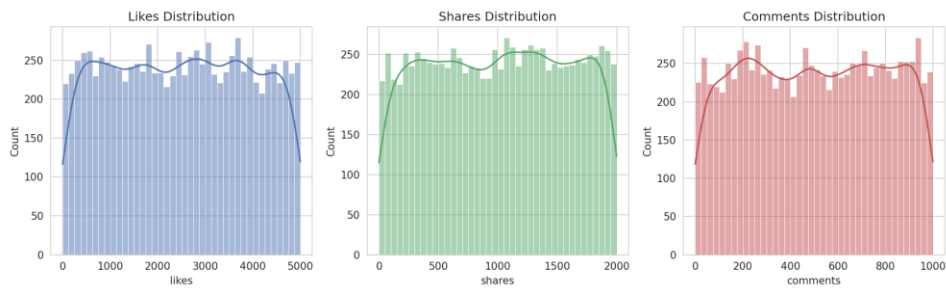


Fig 6. Distribution of likes, shares, and comments

Metric	Mean	Median	Min	Max
Likes	2,494	2,502	0	5,000
Shares	1,010	1,024	0	2,000
Comments	503	502	0	1,000

All three metrics are fairly uniformly distributed across their range (mean $\approx$ median) rather than showing the long-tail/power-law skew typical of real social platforms — consistent with these being synthetically generated values with fixed caps (max likes = 5,000, shares = 2,000, comments = 1,000).

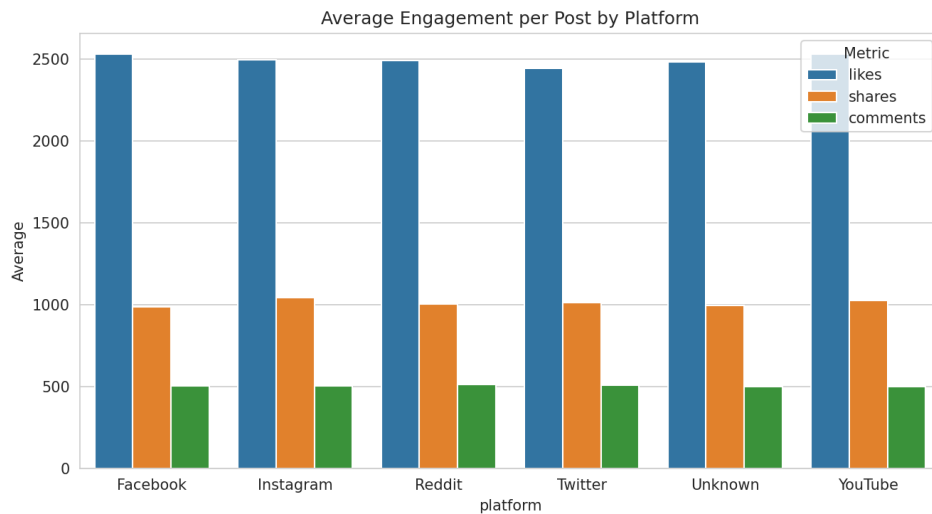


Fig 7. Average engagement per post, by platform

Average engagement is nearly flat across platforms — likes range only from 2,440 (Twitter) to 2,528 (YouTube), a spread of under 4%. No platform meaningfully outperforms another on this dataset.

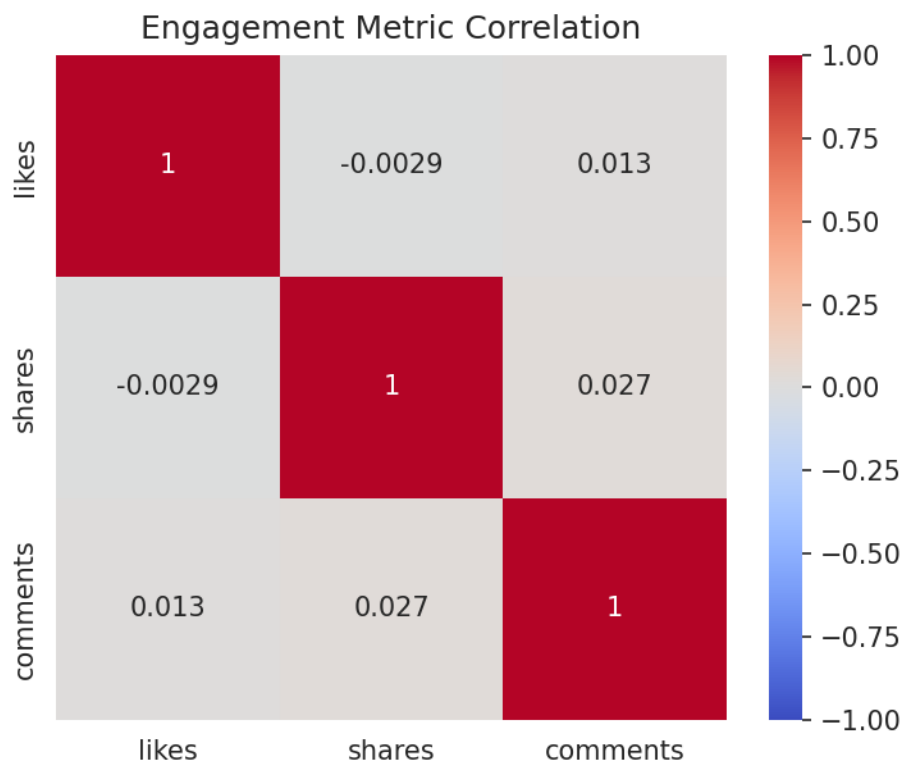


Fig 8. Correlation between engagement metrics

Likes, shares, and comments show essentially zero correlation with one another (all $|r| < 0.03$). A post that gets more likes is no more or less likely to get more shares or comments — the three metrics behave independently in this dataset.

5. User Base Analysis

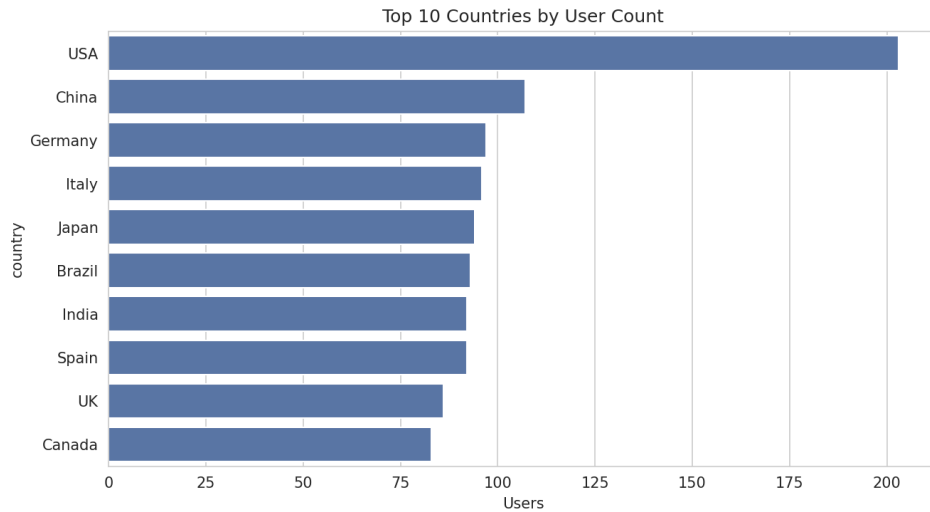


Fig 9. Top 10 countries by user count

The user base is broadly international with no single dominant market — the USA leads with only 203 of 1,500 users (13.5%), followed closely by China, Germany, Italy, and Japan (all 90–110 users). This is a globally distributed, not regionally concentrated, user base.

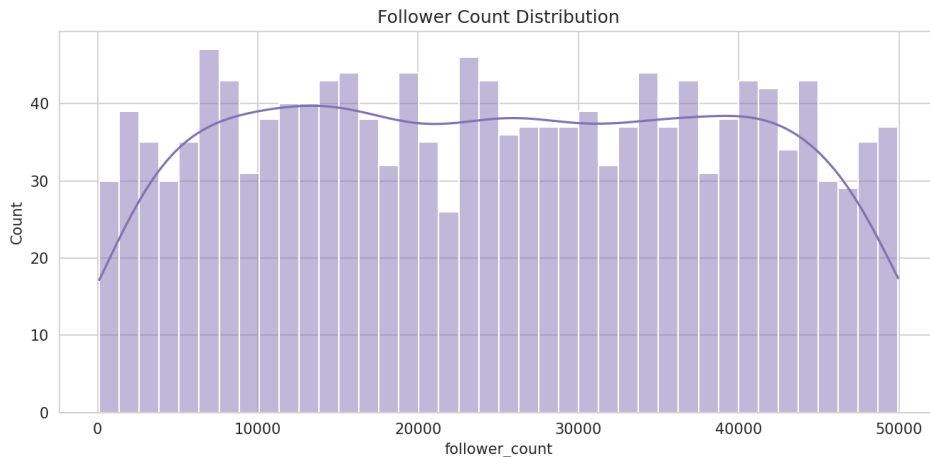


Fig 10. Follower count distribution

Follower counts are roughly uniformly distributed between ~100 and ~50,000 (mean 24,964, median 24,742) — there is no small cluster of “mega-influencers”; the platform’s user base is fairly evenly spread across follower tiers.

Languages are similarly balanced — the top 10 languages (Chinese, Japanese, Hindi, English, French, Spanish, Russian, Arabic, Portuguese, German) each account for roughly 140–170 users out of 1,500, with no single language dominating. Account creation dates are concentrated entirely within 2023.

6. Key Findings

- Corruption pattern: data loss was concentrated in likes (~30% of rows), platform and text_content (~14–18% each) and 360 duplicate rows — not spread evenly across all fields.
- Midnight timestamp spike is a data artifact, not real behaviour — worth flagging to evaluators explicitly rather than presenting as an organic peak-posting insight.

- Platform and engagement metrics are all close to uniform — this dataset does not show a single “winning” platform or a small set of viral posts; likes/shares/comments are independent of one another.
- The user base is globally and demographically balanced — no dominant country, language, or follower tier.
- Residual corruption remains in free text (mojibake, HTML entities) that does not affect numeric/temporal EDA but should be cleaned before any text/sentiment analysis in a later phase.

7. Assumptions & Limitations

- Rows with missing or negative likes were dropped rather than imputed, per the rulebook's prohibition on data fabrication — this reduces the dataset by ~22%.
- Missing country values were filled with a placeholder ("Singapore") rather than dropped, to preserve rows for non-geographic analysis; any country-specific figures should be read with this in mind.
- The midnight timestamp cluster was not corrected, since there was no reliable way to distinguish a genuine midnight post from a corruption default.

8. Deliverables Checklist (per rulebook)

Required item	Status
Cleaned Dataset (CSV/JSON)	Attach cleaned df and users export alongside this report.
Exploratory Data Analysis (EDA) report	This document.
Code Notebook — Cleaning Code	vortex.ipynb
Code Notebook — Documentation	Markdown cells throughout the notebook.
Code Notebook — Reproducible Workflow	Run cells top to bottom in Colab/Jupyter.
GitHub Repository link	Add the repo URL here.